

Auto changeover scheme

What is Auto Change Over?

- Load can be transfer from one Bus to another Bus by using Auto Change Over Scheme.
- It means, within specific conditions, load can be transferred on another bus by automatically ON or OFF the breakers, without disturbing any load.

Different Mode for changeover

- Independent Mode
- Manual Mode / Synchronizing Mode
- Auto Mode
- Blackout Condition

Different Switches for changeover

- Mode Selector Switch:-
- This switch is on Bus coupler.
- This switch is 3 pole, 3 position switch.

Three positions are,

- (1) Auto
- (2) Independent
- (3) Manual

Trip Selector Switch:-

- This switch is on Bus coupler.
- This switch is 3 pole, 3 position switch.

Three positions are,

- (1) Incomer-1 trip
- (2) Incomer-2 trip
- (3) Bus coupler trip

Independent Mode,Manual Mode,Auto Mode,Feeder Required

In this mode, Two Breakers can close at the same time.

For e.g.:-

INCOMER-1	ON
INCOMER-2	ON
BUSCOUPLER	OFF
INCOMER-1	ON
INCOMER-2	OFF
BUSCOUPLER	ON
INCOMER-1	OFF
INCOMER-2	ON
BUSCOUPLER	ON

Manual Mode

- In this mode, When supply is Synchronize, Then Three Breakers can close at the same time. But after some time delay, one breaker will be Trip by using Trip Selector Switch.

For e.g.:-

I/C-1 & B/C are ON & if I/C-2 ON than I/C-1 will be trip after some time delay. (if Trip selector switch is on I/C-1)

Auto Mode

- This mode is only for Bus coupler.
- If two incomer is On & Bus coupler is Off. If one incomer trip due to under voltage. At that time B/C is On automatically.

Feeder Required

- Incomer-1 with Line Pt-1
- Incomer-2 with Line Pt-2
- Bus Coupler
- Bus Pt-1
- Bus Pt-2
- Outgoing Feeders

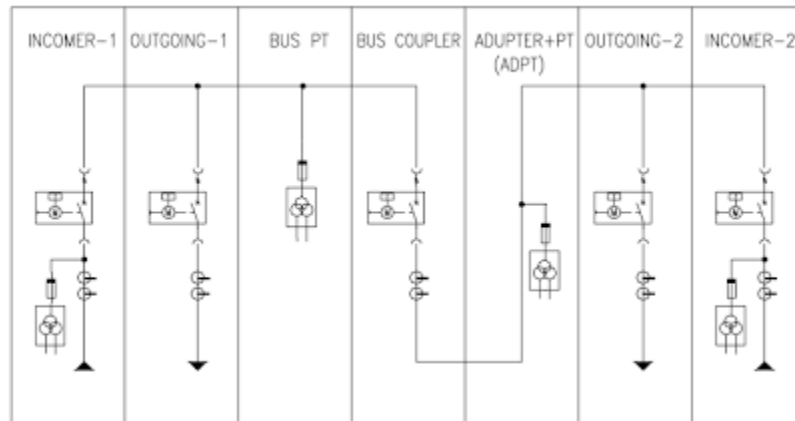
General Requirement,

General Requirement

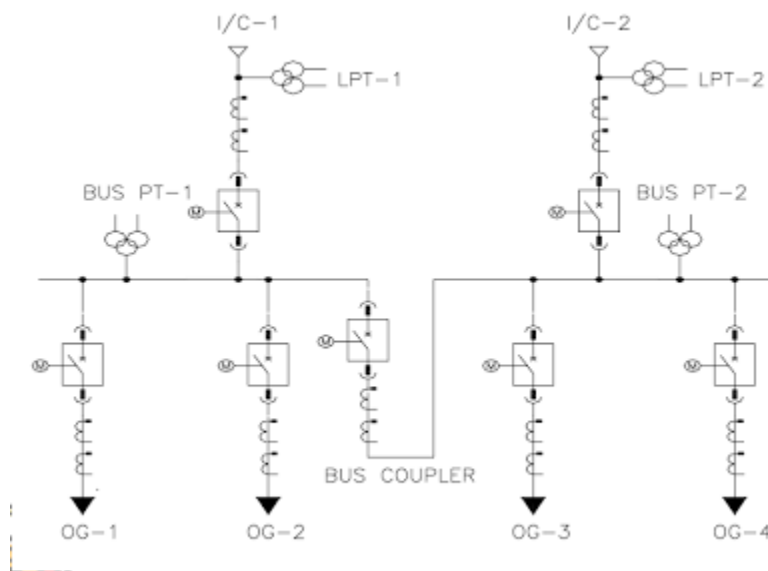
Bus Pt for measuring Bus voltage & checking bus is healthy or unhealthy.

Line Pt for measuring Line voltage & Synchronizing between two sources.
 Numerical relays with maximum BI-BO & breaker with maximum aux. contacts.
 If synchronizing is required, then synchronizing function should be available in numerical relay of bus coupler.

General Schematic Diagram



General Single Line Diagram

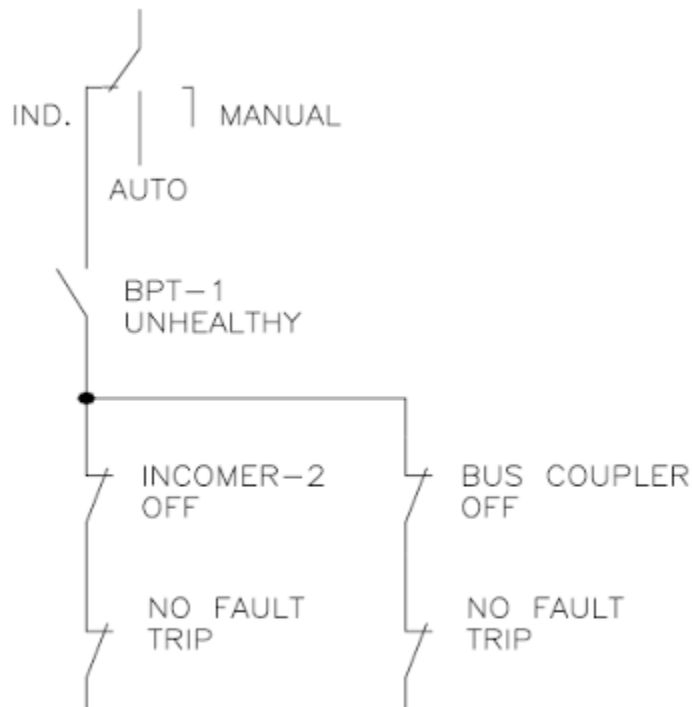


Independent Mode Closing conditions-I/C-1

Mode selector switch in independent mode
Bus PT-1 Unhealthy

Incomer-2 Off & no fault trip
Bus coupler Off & no fault trip

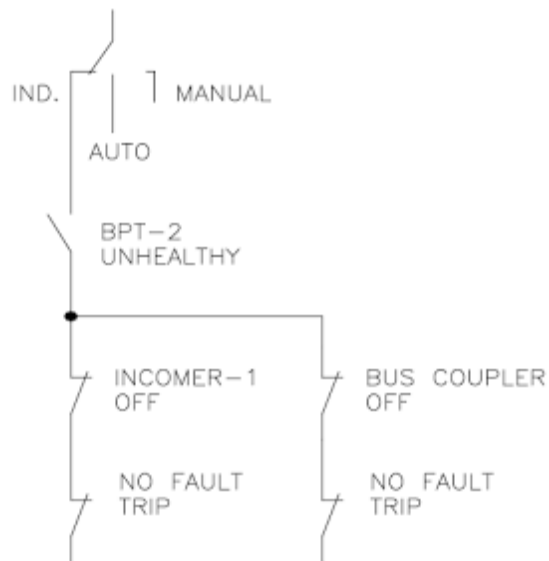
Closing For incomer-1 Ind. mode



Independent Mode

- Mode selector switch in independent mode
- Bus PT-2 Unhealthy
- Incomer-1 Off & no fault trip
- Bus coupler Off & no fault trip

Closing For incomer-2-Ind. mode

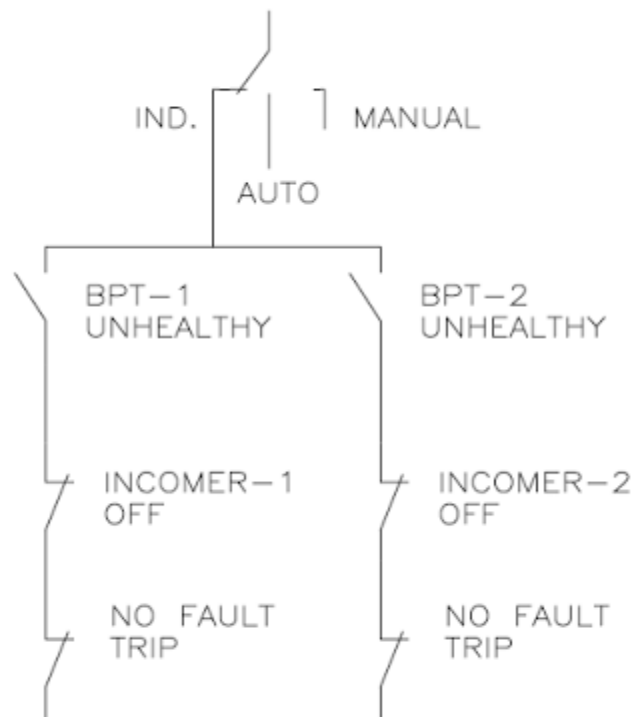


Independent Mode

Closing conditions-B/C

- Mode selector switch in independent mode
- Bus PT-1 Unhealthy
- Bus PT-2 Unhealthy
- Incomer-1 Off & no fault trip
- Incomer-2 Off & no fault trip

For Bus Coupler-Ind. mode

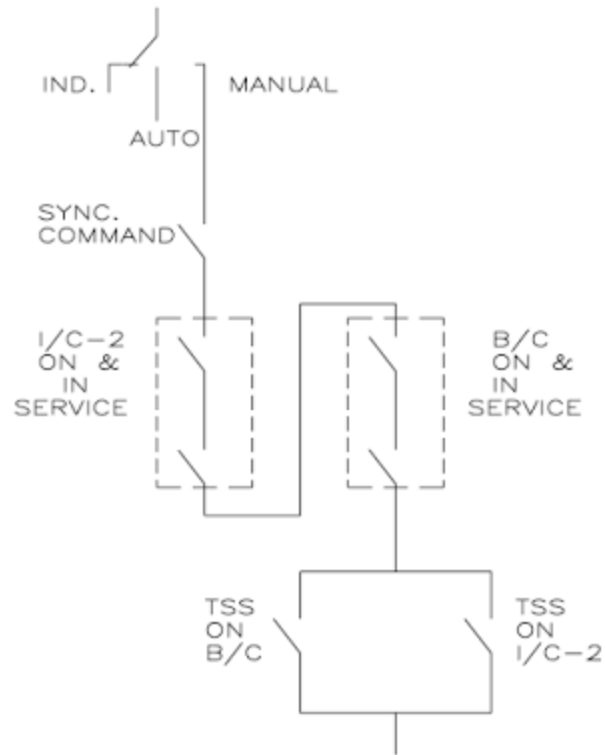


Manual Mode

Manual Mode

Closing conditions-I/C-1

- Mode selector switch in Manual mode
- Supply is Synchronize
- Incomer-2 On & in Service
- Bus coupler On & in Service
- Trip selector switch on B/C
- Trip selector switch on I/C-2

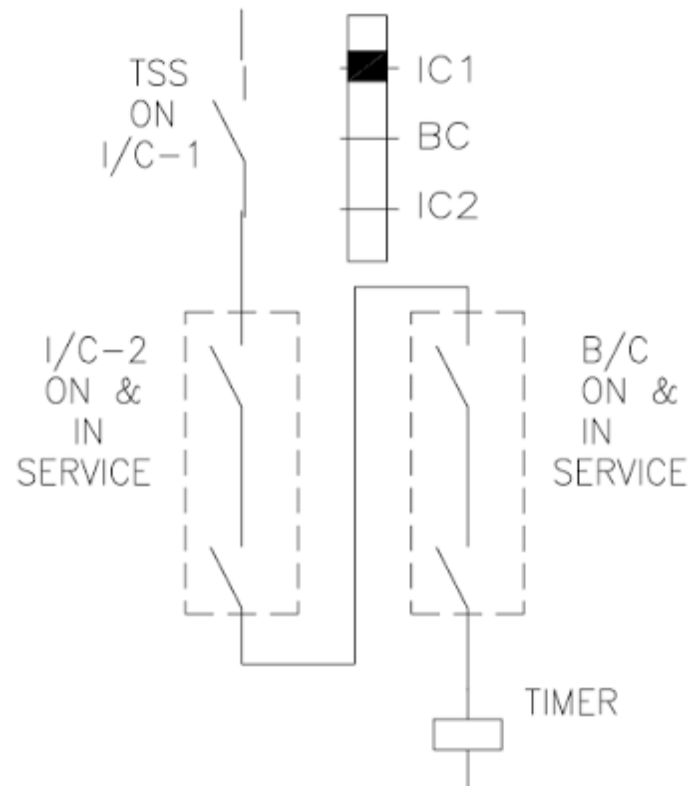


Closing For I/C-1-Manual mode

Manual Mode

Tripping conditions-I/C-1

- Trip selector switch on I/C-1
- Incomer-2 On & in Service
- Bus coupler On & in Service
- After some time delay, I/C-1 will be trip.

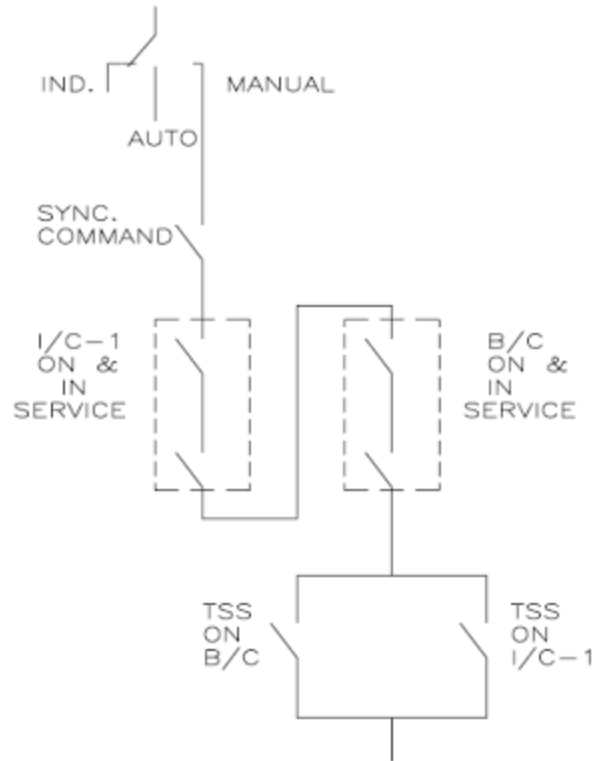


Tripping For I/C-1-Manual mode

Manual Mode

Closing conditions-I/C-2

- Mode selector switch in Manual mode
- Supply is Synchronize
- Incomer-1 On & in Service
- Bus coupler On & in Service
- Trip selector switch on B/C
- Trip selector switch on I/C-1

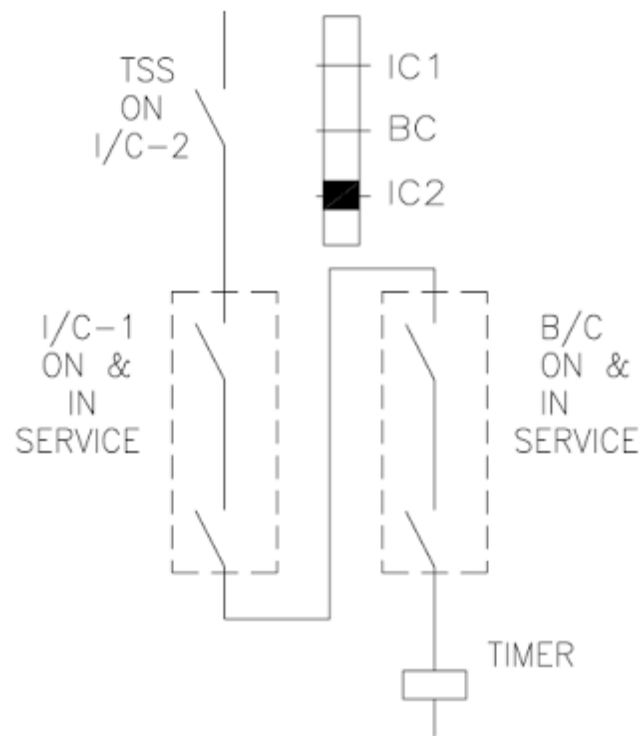


Closing For I/C-2-Manual mode

Manual Mode Tripping conditions

Manual Mode Tripping conditions-I/C-2

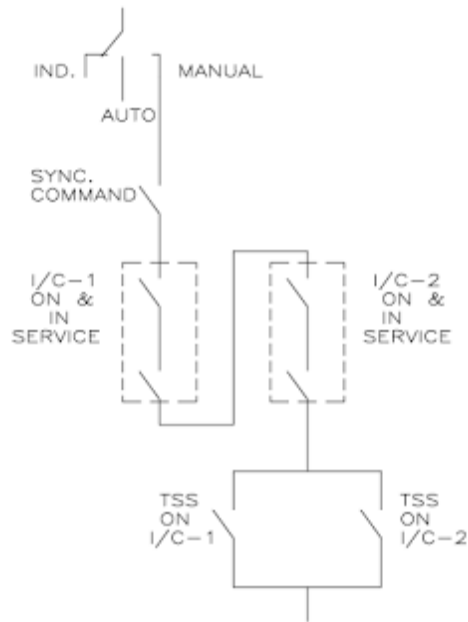
- Trip selector switch on I/C-2
- Incomer-1 On & in Service
- Bus coupler On & in Service
- After some time delay, I/C-2 will be trip.



Tripping For I/C-2-Manual mode

Manual Mode Closing conditions-B/C

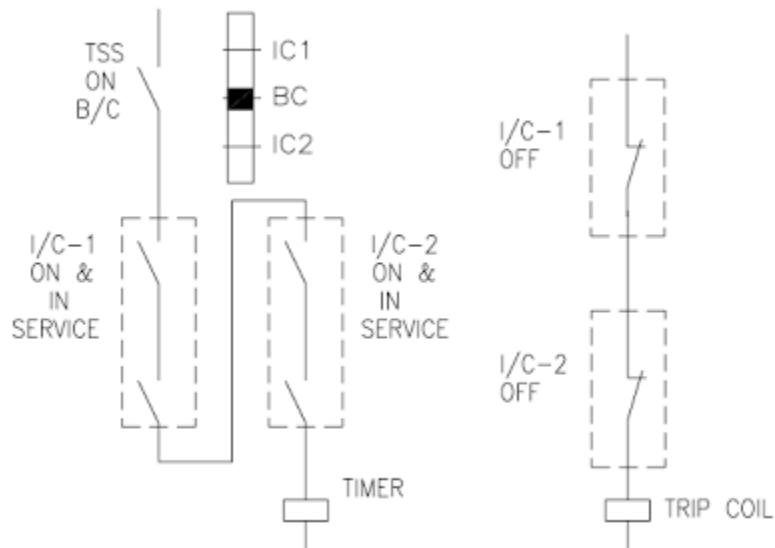
- Mode selector switch in Manual mode
- Supply is Synchronize
- Incomer-1 On & in Service
- Incomer-2 On & in Service
- Trip selector switch on I/C-1
- Trip selector switch on I/C-2



Closing For B/C-Manual mode

Manual Mode Tripping conditions-B/C

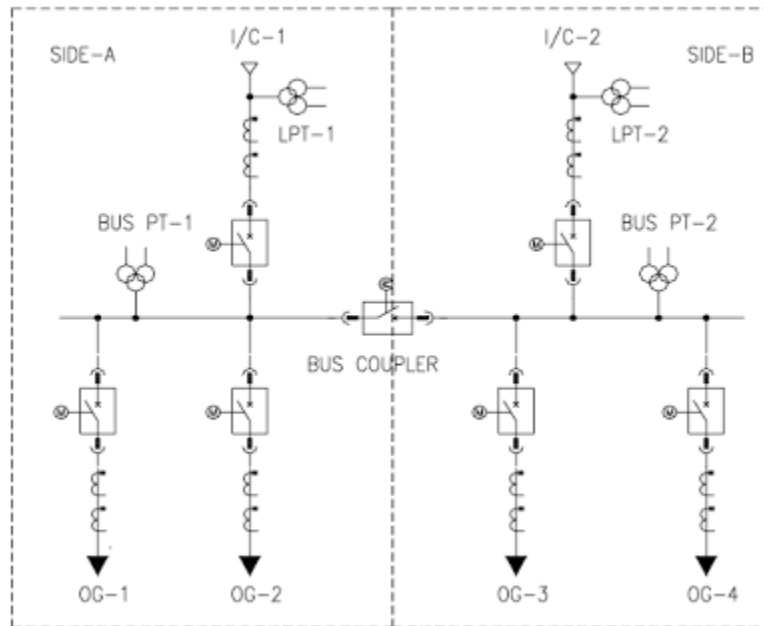
- Trip selector switch on B/C
- Incomer-1 On & in Service
- Incomer-2 On & in Service
- After some time delay, I/C-2 will be trip.
- If in any condition I/C-1 & I/C-2 become OFF, Then B/C will be OFF



Tripping For B/C-Manual mode

Auto Mode

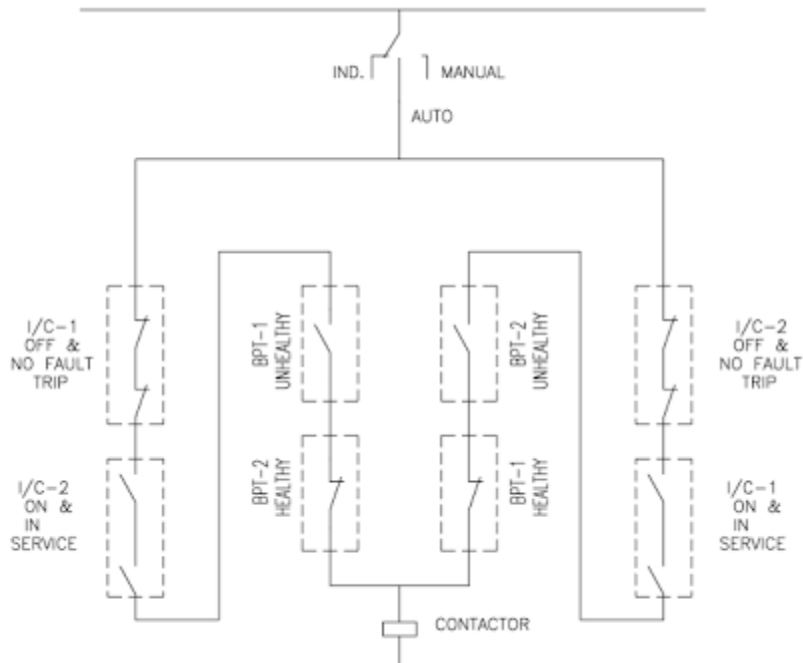
Closing conditions-B/C



Closing conditions-B/C

- When fault (U/V) on bus side-A Than it checks,
 - Selector switch is on Auto Mode.
 - Incomer-1 off & no fault trip.
 - Incomer-2 On & in service.
 - Bus PT-1 unhealthy.
 - Bus PT-2 healthy.
 - This conditions are satisfy than contactor will be pick up & close the breaker.
-
- When fault (U/V) on bus side-B Than it checks,
 - Selector switch is on Auto Mode.
 - Incomer-2 off & no fault trip.
 - Incomer-1 On & in service.
 - Bus PT-2 unhealthy.
 - Bus PT-1 healthy.
 - This conditions are satisfy than contactor will be pick up & close the breaker.

Closing For B/C-Manual mode



Closing For B/C-Manual mode

Additional Points.

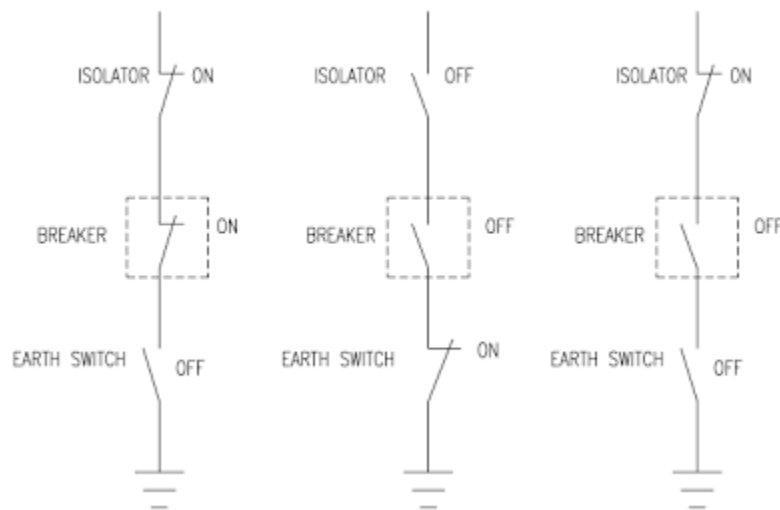
Additional Points.

- If Earth switch or Cable earthing truck & Bus earthing truck is available, than some additional interlocks shall be provided.
- If 4 No Incomer feeders & 1 No Bus coupler are available, than additional interlocks shall be provided

Additional Interlocks

- It will not be possible to OPEN the front DOOR when the circuit breaker is switched "ON" in service position.
- It will not be possible to rack "OUT" the Vacuum Circuit Breaker trolley from Service to Test position when the breaker is switched "ON"
- Similarly, it will not be possible to rack "IN" the Vacuum Circuit Breaker from Test to Service position when Vacuum Circuit Breaker is switched "ON".
- It will not be possible to rack "IN" the Vacuum Circuit Breaker when the front Vacuum Circuit Breaker chamber door is open. (However, a suitable defeat interlock mechanism shall be provided for emergency purpose.)
- It will be only possible to switch "ON" the earthling switch if provided, when the withdrawable truck is in Test position.

- It will not be possible to rack “IN” Vacuum Circuit Breaker from Test to service position, in case the earthing switch is “ON”.
- Earthing trolleys can be also supplied. Electro-magnetic interlocks to be provided on earthing trolley for incomers and ties feeders.
- It will be not possible to switch “ON” the circuit breaker when ISOLATOR is Open.
- It will be not possible to OPEN the isolator when circuit breaker is ON.
- It will be not possible to “CLOSE” the earth switch when circuit breaker is “ON”.
- It will be not possible to “CLOSE” the earth switch when isolator is “CLOSED”.
- Similarly, it will be not possible to “CLOSE” the circuit breaker & isolator when earth switch is “ON”.



Schematic diagram of interlocks